

101. Let z and w be two complex numbers such that $w = z\bar{z} - 2z + 2$, $\left| \frac{z+i}{z-3i} \right| = 1$ and $\text{Re}(w)$ has minimum value.

Then, the minimum value of $n \in \mathbb{N}$ for which w^n is real, is equal to _____.

माना दो सम्मिश्र संख्याओं z तथा w के लिए $w = z\bar{z} - 2z + 2$, $\left| \frac{z+i}{z-3i} \right| = 1$ हैं, तथा $\text{Re}(w)$ का मान निम्नतम है। तो $n \in \mathbb{N}$ का निम्नतम मान, जिसके लिए w^n वास्तविक है, बराबर है _____।

Ans. 4

Sol. $z = x + iy$

$$\left| \frac{z+i}{z-3i} \right| = 1$$

$$|z+i| = |z-3i|$$

$$|x+iy+i| = |x+y-3i|$$

$$\Rightarrow y = 1$$

$$w = z\bar{z} - 2z + 2$$

$$w = x^2 - 2x + 3 - 2i$$

$$\text{Re}(w) = x^2 - 2x + 3 \text{ will be minimum when } x = 1$$

$$\Rightarrow w = 2 - 2i = 2\sqrt{2} e^{-i\frac{\pi}{4}}$$

$$w^n = (2\sqrt{2})^n e^{-i\frac{n\pi}{4}}$$

$$\Rightarrow \frac{n\pi}{4} = k\pi$$

$$n = 4k$$

$$n|_{\min} = 4$$

102. Four fair dice are thrown independently 27 times. Then the expected number of times, at least two dice show up a three or a five, is _____.

चार अनभिन्नत पासों को 27 बार स्वतंत्र रूप से फेंका जाता है। तो कम से कम दो पासों के तीन या पांच दर्शाने की संभावना कितनी बार है?

Ans. 11

Sol. $P(\text{at least two are 3 or 5}) = 1 - P(\text{No 3 or 5}) - P(\text{Exactly one 3 or 5})$

$$= 1 - \left(\frac{4}{6}\right)^4 - {}^4C_1 \frac{2}{6} \left(\frac{4}{6}\right)^3$$

$$= 1 - \frac{16}{81} - \frac{32}{81} = \frac{11}{27}$$

In binomial distribution, $n = 27$, $p = \frac{11}{27}$, $q = \frac{16}{27}$

Expected value $= np$

$$= 27 \times \frac{11}{27} = 11$$

103. The remainder when 428^{2024} is divided by 21 is _____.

428^{2024} को 21 से विभाजित करने का शेषफल है _____

Ans. 1

Sol. $21 \times 20 = 420 \Rightarrow (428)^{2024} = (420+8)^{2024}$

$$= {}^{2024}C_0 (420)^{2024} + \dots + {}^{2024}C_{2033} (420)^{2023} + 8^{2024}$$

$$= 420 [\text{some integer}] + (64)^{1012}; \quad 63 = 21 \times 3$$

$$= 21 \times 20 [\text{some integer}] + (63 + 1)^{1012}$$

$$= 21 \times 20 [\text{some integer}] + 63 [\text{some other integer}] + 1$$

Hence remainder is 1.

104. A bag contains six balls of different colours. Two balls are drawn in succession with replacement. The probability that both are of the same colour is p . Next four balls are drawn in succession with replacement and the probability that exactly three balls are of the same colour is q . If $p : q = m : n$, where m and n are coprime, then $m + n$ is equal to _____.

एक थैले में भिन्न रंगों की छः गेंद हैं। माना एक-एक कर प्रतिस्थापन सहित दो गेंद निकाली जाती हैं तथा दोनों गेंदों के एक ही रंग के होने की प्रायिकता p है। फिर एक-एक कर प्रतिस्थापन सहित चार गेंद निकाली जाती हैं तथा ठीक तीन गेंदों का एक ही रंग के होने की प्रायिकता q है। यदि $p : q = m : n$ है, जहाँ m व n सहअभाज्य हैं तब $m + n$ बराबर है।

Ans. 14

Sol. $p = \frac{{}^6C_1}{6 \times 6} = \frac{1}{6}$

$$q = \frac{{}^6C_1 \times {}^5C_1 \times 4}{6 \times 6 \times 6 \times 6} = \frac{5}{54}$$

$$\therefore p : q = 9 : 5 \Rightarrow m + n = 14$$

105. Let $S = \left\{ z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R} \right\}$. If $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$, then $242\alpha^2$ is equal to _____.

माना $S = \left\{ z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R} \right\}$ है। यदि $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$ है, तो $242\alpha^2$ बराबर है _____

Ans. 1680

Sol. $\left(\frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \right) \in \mathbb{R}$

$$\Rightarrow 1 + \frac{(11iz - 13)}{(z^2 - 3iz - 2)} \in \mathbb{R}$$

Put $z = \alpha - \frac{13}{11}i$

$$\Rightarrow (z^2 - 3iz - 2) \text{ is imaginary}$$

Put $z = x + iy$

$$\Rightarrow (x^2 - y^2 + 2xyi - 3ix + 3y - 2) \in \text{Imaginary}$$

$$\Rightarrow \text{Re}(x^2 - y^2 + 3y - 2 + (2xy - 3x)i) = 0$$

$$\Rightarrow x^2 - y^2 + 3y - 2 = 0$$

$$x^2 = y^2 - 3y + 2$$

$$y^2 = (y - 1)(y - 2)$$

$$\therefore z = \alpha - \frac{13}{11}i$$

Put $x = \alpha, y = \frac{-13}{11}$

$$\alpha^2 = \left(\frac{-13}{11} - 1 \right) \left(\frac{-13}{11} - 2 \right)$$

$$\alpha^2 = \frac{(24 \times 35)}{121}$$

$$242\alpha^2 = 48 \times 35 = 1680$$

106. If the mean of the frequency distribution

Class :	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Frequency :	2	3	x	5	4

is 28, then its variance is _____.

यदि बारंबारता बंटन

वर्ग :	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
बारंबारता :	2	3	x	5	4

का माध्य 28 है, तो इसका प्रसरण है _____

Ans. 151

Sol. Mean = 28 = $\frac{5 \times 2 + 15 \times 3 + 25 \times x + 35 \times 5 + 45 \times 4}{14 + x}$

$$392 + 28x = 410 + 25x$$

$$3x = 18$$

$$x = 6$$

Now variance

$$= \frac{\sum f_i (x_i)^2}{\sum f_i} - (\bar{x})^2$$

$$= \frac{2(25) + 3(15)^2 + 6(25)^2 + 5(35)^2 + 4(45)^2}{20} - 28^2$$

$$= \frac{50 + 675 + 3750 + 6125 + 8100}{20} - 784$$

$$= \frac{18700}{20} - 784 = 935 - 784$$

$$= 151$$

107. Let the mean of the data

x	1	3	5	7	9
Frequency (f)	4	24	28	α	8

be 5. If m and σ^2 are respectively the mean deviation about the mean and the variance of the data, then

$\frac{3\alpha}{m + \sigma^2}$ is equal to _____.

माना आंकड़ों

x	1	3	5	7	9
बारंबारता (f)	4	24	28	α	8

का माध्य 5 है। यदि इन आंकड़ों के माध्य के सापेक्ष माध्य विचलन तथा प्रसरण क्रमशः m तथा σ^2 हैं, तो $\frac{3\alpha}{m + \sigma^2}$ बराबर है _____.

Ans. 8

Sol.
$$\frac{4 + 72 + 28 \times 5 + 7\alpha + 72}{64 + \alpha} = 5$$

$$\Rightarrow \alpha = 16$$

$$\sum f_i = 80$$

$$M.D = \frac{\sum f_i |x_i - 5|}{\sum f_i}$$

$$= \frac{4 + 4 + 24 \times 2 + 0 + 16 \times 2 + 8 \times 4}{80}$$

$$= \frac{8}{5}$$

$$\sigma^2 = \frac{\sum f_i |x_i - 5|}{\sum f_i}$$

$$= \frac{4 + 16 + 24 \times 4 + 0 + 16 \times 4 + 8 \times 16}{80} = \frac{22}{5}$$

$$\frac{3\alpha}{m + \sigma^2} = \frac{3 \times 16}{\frac{8}{5} + \frac{22}{5}} = 8$$

108. Let $X = \{11, 12, 13, \dots, 40, 41\}$ and $Y = \{61, 62, 63, \dots, 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $|\bar{x} + \bar{y} - \sigma^2|$ is equal to _____.

माना प्रेक्षणों के दो समुच्चय $X = \{11, 12, 13, \dots, 40, 41\}$ तथा $Y = \{61, 62, 63, \dots, 90, 91\}$ है। यदि इनके माध्य क्रमशः $\bar{x}$ तथा $\bar{y}$ हैं तथा $X \cup Y$ में सभी प्रेक्षणों का प्रसरण σ^2 है, तो $|\bar{x} + \bar{y} - \sigma^2|$ बराबर है _____

Ans. 603

Sol. $\bar{x} = \frac{\sum_{i=11}^{41} i}{31} = \frac{11+41}{2} = 26$ (31 elements)

$$\bar{y} = \frac{\sum_{j=61}^{91} j}{31} = \frac{61+91}{2} = 76 \text{ (31 elements)}$$

Combined mean, $\mu = \frac{31 \times 26 + 31 \times 76}{31 + 31}$

$$= \frac{26 + 76}{2} = 51$$

$$\sigma^2 = \frac{1}{62} \times \left(\sum_{i=1}^{31} (x_i - \mu)^2 + \sum_{i=1}^{31} (y_i - \mu)^2 \right) = 705$$

Since, $x_i \in X$ are in A.P. with 31 elements & common difference 1, same is $y_i \in y$, when written in increasing order.

$$\therefore \sum_{i=1}^{31} (x_i - \mu)^2 = \sum_{i=1}^{31} (y_i - \mu)^2$$

$$= 10^2 + 11^2 + \dots + 40^2$$

$$= \frac{40 \times 41 \times 81}{6} - \frac{9 \times 10 \times 19}{6} = 21855$$

$$\therefore |\bar{x} + \bar{y} - \sigma^2| = |26 + 76 - 705| = 603$$

109. The number of seven digit positive integers formed using the digits 1, 2, 3 and 4 only and sum of the digits equal to 12 is _____.

केवल अंको 1, 2, 3 तथा 4 के प्रयोग से बनने वाले सात अंकों के धनात्मक पूर्णांकों, जिनके अंकों का योग 12 है, की संख्या है

Ans. 413

Sol. $x_1 + x_2 + x_3 + \dots + x_7 = 12$

Number of solutions

$$= \text{Coefficient of } x^{12} \text{ in } (x^1 + x^2 + x^3 + x^4)^7$$

$$= \text{Coefficient of } x^5 \text{ in } (1 + x + x^2 + x^3)^7$$

$$= \text{Coefficient of } x^5 \text{ in } (1-x^4)^7 (1-x)^{-7}$$

$$= \text{Coefficient of } x^5 \text{ in } (1-7x^4)(1-x)^{-7}$$

$$= \text{Coefficient of } x^5 \text{ in } (1-7x^4) \sum_{r=0}^{\infty} {}^{7+r-1}C_r \cdot x^r$$

$$= {}^{11}C_5 - 7 \times {}^7C_1$$

$$= 462 - 49 = 413$$

110. If $1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \dots$ upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right) \log_e \left(\frac{a}{b}\right)$, where a and b are integers with $\gcd(a, b) = 1$, then $11a + 18b$ is equal to _____.

$$\text{यदि } 1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \dots \text{ upto } \infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right) \log_e \left(\frac{a}{b}\right), \text{ है, जहाँ}$$

a तथा b पूर्णांक हैं तथा $\gcd(a, b) = 1$ है, तो $11a + 18b$ बराबर है _____

Ans. 76

Sol.
$$S = 1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \dots \infty$$

$$= 1 + \frac{(1-\sqrt{2}/\sqrt{3})}{2} + \frac{(1-\sqrt{2}/\sqrt{3})^2}{6} + \frac{(1-\sqrt{2}/\sqrt{3})^3}{12} + \dots \infty$$

$$\text{let } 1 - \frac{\sqrt{2}}{\sqrt{3}} = a$$

$$S = 1 + \frac{a}{2} + \frac{a^2}{6} + \frac{a^3}{12} + \dots$$

$$= 1 + \left(1 - \frac{1}{2}\right)a + \left(\frac{1}{2} - \frac{1}{3}\right)a^2 + \left(\frac{1}{3} - \frac{1}{4}\right)a^3 + \dots$$

$$= 1 + \left(a + \frac{a^2}{2} + \frac{a^3}{3} \dots \infty\right) + \frac{1}{a} \left(\frac{-a^2}{2} - \frac{a^3}{3} - \frac{a^4}{4} \dots \infty\right)$$

$$= -\ln(1-a) + \frac{1}{a} \left(-a - \frac{a^2}{2} - \frac{a^3}{3} \dots \infty\right) + 2$$

$$= -\ln(1-a) + \frac{1}{a} \ln(1-a) + 2$$

$$= 2 + \left(\frac{1}{a} - 1\right) \ln(1-a)$$

$$= 2 + \left(\frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}} - 1\right) \ln\left(1 - 1 + \sqrt{\frac{2}{3}}\right)$$

$$= 2 + \frac{\sqrt{2}}{\sqrt{3}-\sqrt{2}} \ln \sqrt{\frac{2}{3}}$$

$$= 2 + \left(\frac{\sqrt{6}+2}{1} \cdot \frac{1}{2} \ln \frac{2}{3}\right)$$

$$= 2 + \left(\sqrt{\frac{3}{2}} + 1\right) \ln \frac{2}{3}$$

$$\therefore 11a + 18b = 76$$

111. The number of 9 digit numbers, that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is _____.

संख्या 123412341 के सभी अंकों के प्रयोग से बनाई जा सकने वाली 9 अंकों की ऐसी संख्याओं, कि सम अंक केवल सम स्थानों पर हो, की संख्या है।

Ans. 60

Sol. Four even digits at four places in $\frac{4!}{2!2!} = 6$ ways

Five odd digits at odd places = $\frac{5!}{3!2!} = 10$ ways

Required number of numbers = 60 ways.

112. If $\sum_{r=0}^5 \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$, $\gcd(m, n) = 1$, then $m - n$ is equal to _____.

यदि $\sum_{r=0}^5 \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$, $\gcd(m, n) = 1$ है, तो $m - n$ बराबर है _____.

Ans. 2035

Sol.
$$\frac{1}{12} \sum_{r=0}^5 \frac{12}{2r+2} {}^{11}C_{2r+1} = \frac{1}{12} \sum_{r=0}^5 {}^{12}C_{2r+2}$$

$$= \frac{1}{12} [{}^{12}C_2 + {}^{12}C_4 + {}^{12}C_6 + {}^{12}C_8 + {}^{12}C_{10} + {}^{12}C_{12}]$$

$$= \frac{1}{12} [2^{12-1} - 1] = \frac{2047}{12}$$

$$\Rightarrow m - n = 2047 - 12 = 2035$$

113. Suppose Anil's mother wants to give 5 whole fruits to Anil from a basket of 7 red apples, 5 white apples and 8 oranges. If in the selected 5 fruits, at least 2 oranges, at least 2 oranges, at least one red apple and at least one white apple must be given, then the numbers of ways, Anil's mother can offer 5 fruits to Anil is _____.

माना अनिल की माँ एक टोकरी, जिसमें 7 लाल सेब, 5 सफेद सेब तथा 8 संतरे हैं, में से 5 फल अनिल को देना चाहती है। यदि टोकरी से लिए गए 5 फलों में कम से कम 2 संतरे, कम से कम एक लाल सेब तथा कम से कम एक सफेद सेब अवश्य होने चाहिए, तो अनिल की माँ द्वारा अनिल को 5 फल देने के तरीकों की संख्या है।

Ans. 6860 or 3

Sol.

	7 Red Apple	5 white Apple	8 oranges
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$$x + y + z = 5$$

$$\geq 1 \quad \geq 1 \quad \geq 2$$

C-I 2 1 2

C-II 1 2 2

C-III 1 1 3

$${}^7C_2 \cdot {}^5C_1 \cdot {}^8C_2 + {}^7C_1 \cdot {}^5C_2 \cdot {}^8C_2 + {}^7C_1 \cdot {}^5C_1 \cdot {}^8C_3$$

$$= 2940 + 1960 + 1960 = 6860$$

114. Number of 4-digit numbers (the repetition of digits is allowed) which are made using the digits 1,2,3 and 5, and are divisible by 15, is equal to _____.

अंकों 1,2,3 तथा 5 के प्रयोग से (अंकों की पुनरावृत्ति की अनुमति है) 15 से विभाज्य 4 अंकों की संख्याओं की संख्या है।

Ans. 21

Sol. divisible by 15,

In which last digit should be 5 and sum of digits must be divisible by 3

$$C-I : \underline{1} \underline{2} \underline{1} (\underline{5}) \rightarrow \frac{3!}{2!} = 3$$

$$\underline{2} \underline{2} \underline{3} (\underline{5}) \rightarrow \frac{3!}{2!} = 3$$

$$\underline{3} \underline{3} \underline{1} (\underline{5}) \rightarrow \frac{3!}{2!} = 3$$

$$\underline{1} \underline{1} \underline{5} (\underline{5}) \rightarrow \frac{3!}{2!} = 3$$

$$\underline{2} \underline{3} \underline{5} (\underline{5}) \rightarrow 3! = 6$$

$$\underline{3} \underline{5} \underline{5} (\underline{5}) \rightarrow \frac{3!}{2!} = 3$$

Total = 21

115. The sum of all the four-digit numbers that can be formed using all the digits 2, 1, 2, 3 is equal to _____.

सभी अंकों 2, 1, 2, 3 के प्रयोग से 4 अंकों की बनाई जा सकने वाली सभी संख्याओं का योग है :

Ans. 26664

Sol. Sum of all 4 digit numbers

= sum of digit at unit place $\times 10^0$

+ sum of digit at term place $\times 10^1$

+ sum of digit at hundred place $\times 10^2$

+ sum of digit at thousand place $\times 10^3$

$$= \left(1 \times \frac{3!}{2!} + 2 \times 3! + 3 \times \frac{3!}{2!} \right) (10^0 + 10^1 + 10^2 + 10^3)$$

$$= 24 \times (1111)$$

$$= 26664$$

116. Five digit numbers are formed using the digits 1,2,3,5,7 with repetitions and are written in descending order with serial numbers. For example, the number 77777 has serial number 1. Then the serial number of 35337 is _____.

अंकों 1,2,3,5,7 के प्रयोग से, पुनरावृत्ति के साथ पाँच अंकों की संख्याएँ बनाई जाती हैं तथा इनको ह्रसमान क्रम में, क्रम संख्या के साथ लिखा जाता है। उदाहरण के लिए संख्या 77777 की क्रम संख्या 1 है। तब 35337 की क्रम संख्या है।

Ans. 1436

Sol. Number of numbers starting with 7 $\rightarrow$ 625
 Number of numbers starting with 5 $\rightarrow$ 625
 Number of numbers starting with 37 $\rightarrow$ 125
 Number of numbers starting with 357 $\rightarrow$ 25
 Number of numbers starting with 3537 $\rightarrow$ 5
 Number of numbers starting with 3535 $\rightarrow$ 5
 Number of numbers starting with 35337 $\rightarrow$ 1

1436

The position of the number 35337 is 1436.

117. If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is _____.

यदि बारंबारता बंटन

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

के माध्य तथा प्रसरण क्रमशः 9 तथा 15.08 है, तो $\alpha^2 + \beta^2 - \alpha\beta$ का मान है _____

Ans. 25

Sol. Mean = $\frac{8 + 16 + 120 + 80 + 56 + 80 + 6\alpha + 12\beta}{40 + \alpha + \beta}$

$$\Rightarrow 360 + 9\alpha + 9\beta = 360 + 6\alpha + 12\beta$$

OR

$$3\alpha - 3\beta = 0$$

$$16 + 64 + 36\alpha + 960 + 800$$

$$15.08 + 81 = \frac{+144\alpha + 784 + 1280}{40 + 2\alpha}$$

$$(40 + 2\alpha)(96.08) = 3904 + 180\alpha$$

$$3843.20 + (192.16)\alpha = 3904 + 180\alpha$$

$$(12.16)\alpha = 60.80$$

$$\alpha = 5 = \beta$$

118. If the least and the largest real values of α , for which the equation $z + \alpha |z - 1| + 2i = 0$ ($z \in \mathbb{C}$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to _____.

यदि α के न्यूनतम तथा अधिकतम वास्तविक मान, जिनके समीकरण $z + \alpha |z - 1| + 2i = 0$ ($z \in \mathbb{C}$ तथा $i = \sqrt{-1}$) का हल है, क्रमशः p तथा q हैं, तो $4(p^2 + q^2)$ बराबर है _____।

Ans. 10

Sol. $z = x + iy$

$$x + \alpha \sqrt{(x-1)^2 + y^2} + i(y+2) = 0 + 0i$$

$$y = -2$$

$$x + \alpha \sqrt{(x-1)^2 + y^2} = 0$$

$$x^2 = \alpha^2 ((x-1)^2 + 4)$$

$$\alpha^2 = \frac{x^2}{x^2 - 2x + 5} = f(x)$$

$$f'(x) = \frac{-2x(x-5)}{(x^2 - 2x + 5)^2}$$

$$f'(x) = \frac{(-)ve}{0} \quad (+)ve \quad (-)ve$$

$$\lim_{x \rightarrow \pm\infty} f(x) = 1$$

$$f(0) = 0$$

$$f(5) = \frac{5}{4}$$

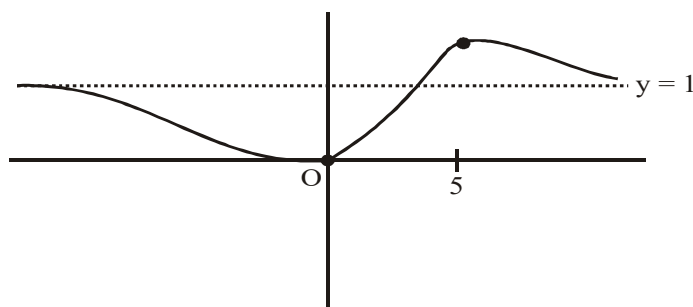
$$\alpha^2 \in \left[0, \frac{5}{4}\right]$$

$$\alpha \in \left[\frac{-\sqrt{5}}{2}, \frac{\sqrt{5}}{2}\right]$$

$$p = \frac{-\sqrt{5}}{2}$$

$$q = \frac{\sqrt{5}}{2}$$

$$4(p^2 + q^2) = 10$$



119. The students $S_1, S_2, \dots, S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____.

छात्रों, $S_1, S_2, \dots, S_{10}$ को तीन समूहों A, B तथा C में इस प्रकार विभाजित करना है कि प्रत्येक समूह में कम से कम एक छात्र हो तथा समूह C में अधिक 3 छात्र हों। तो इस प्रकार समूह बनाने की कुल संभावनायें हैं _____।

Ans. 31650

Sol. Case – I Group C has 1 student

$$\text{number of ways} = {}^{10}C_1 (2^9 - 2) = A_1 = 5100$$

Case – II Group C has 2 student

$$\text{number of ways} = {}^{10}C_2 (2^8 - 2) = A_2 = 11430$$

Case – III Group C has 3 student

$$\text{number of ways} = {}^{10}C_3 (2^7 - 2) = A_3 = 15120$$

$$\text{Ans.} = A_1 + A_2 + A_3 = 31650$$

120. Let $\alpha = \sum_{k=0}^n \left(\frac{{}^nC_k}{k+1} \right)^2$ and $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^nC_k {}^nC_{k+1}}{k+2} \right)$. If $5\alpha = 6\beta$, then n equals _____.

$$\text{माना } \alpha = \sum_{k=0}^n \left(\frac{{}^nC_k}{k+1} \right)^2 \text{ तथा } \beta = \sum_{k=0}^{n-1} \left(\frac{{}^nC_k {}^nC_{k+1}}{k+2} \right) \text{ हैं। यदि } 5\alpha = 6\beta \text{ हैं, तो n बराबर है } \underline{\hspace{2cm}}$$

Ans. 10

$$\text{Sol. } \alpha = \sum_{k=0}^n \frac{{}^nC_k \cdot {}^nC_k}{k+1} \cdot \frac{n+1}{n+1}$$

$$= \frac{1}{n+1} \sum_{k=0}^n {}^{n+1}C_{k+1} \cdot {}^nC_{n-k}$$

$$\alpha = \frac{1}{n+1} \cdot {}^{2n+1}C_{n+1}$$

$$\beta = \sum_{k=0}^{n-1} {}^nC_k \cdot \frac{{}^nC_{k+1}}{k+2} \cdot \frac{n+1}{n+1}$$

$$\frac{1}{n+1} \sum_{k=0}^{n-1} {}^nC_{n-k} \cdot {}^{n+1}C_{k+2}$$

$$= \frac{1}{n+1} \cdot {}^{2n+1}C_{n+2}$$

$$\frac{\beta}{\alpha} = \frac{{}^{2n+1}C_{n+2}}{{}^{2n+1}C_{n+1}} = \frac{2n+1-(n+2)+1}{n+2}$$

$$\frac{\beta}{\alpha} = \frac{n}{n+2} = \frac{5}{6}$$

$$n = 10$$